SINKGRAB · DESIGN DECISION RECORD

SinkGrab decisions of 2026-10-25 (ring sinking, new requirements, overload device, scraper crew)

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SKG-DDR-006	Version	0.1
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/decisions/0006-decisions-2026-10-25.md @ ec186d3		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Amish's decisions O2 to O5 of 2026-10-25 recorded and carried out in the model, calculations, STEM check, requirements, risk register, bill of materials, drawings and build plan

0006: Decisions of 2026-10-25

- **Date:** 2026-10-25
- **Status:** accepted

Context

The kit 1.9.1 upgrade and the independent review of the STEM check left five open decisions in the design decisions register (SKG-DEC-001 O1 to O5), each posed as a state, options and a recommendation, and listed in a report to Amish. Amish Chadha (owner), 2026-10-25 12:43 CT: "i approve of your recommendations".

That decides the four open decisions that carried a recommendation (O2 to O5), each as recommended. O1, the concept choice, was sent without a pick because options A and B score within 10 % (STANDARDS section 21), so it stays open and option A stays the design of record (SKG-DDR-005).

Options considered

#	Question	Options	Chosen
O2	Ring sinking (PR-001): on paper the central friction estimate (2 kPa) already exceeds the 1.66 kPa the string can overcome	A: trial as designed and decide after measuring. B: plan a bentonite slurry fed down a pipe into the overcut annulus before the trial and test it in the test pit. C: sixteen saddles	**B**
O3	Two new requirements: PR-012 (air test that reads carbon dioxide) and PR-013 (soils)	A: accept both as written. B: accept PR-012 only	**A**
O4	Overload window and proof basis (PR-007): no shear pin fits between the working pull and the tripod's rating, and the crank pin fatigues	A: keep the shear devices and lower the working load. B: a dedicated or uprated tripod. C: a calibrated friction slip clutch on the crank shaft with a load indicator at the dead end, proof at 1.25 times the rated load	**C** (B stays open for later, if the trials show more bite is worth it)
O5	Scraper turning force: 142 N for each of two people against sustained limits of about 100 N	A: a 2.0 m T-bar if the well-head layout allows it. B: four people, two on each side. C: keep it and assess in the trial	**A if the layout allows it, else B**: the layout does not allow A, so **B**

Decision

1. **O2, option B.** A bentonite slurry, mixed in a drum at the well head and pumped down a feed pipe into the 40 mm annulus the toe blades cut outside the bottom ring, is part of the sinking method from the first trial, as in caisson practice; it is tried in the test pit before a partner well. About 0.45 m3 fills the annulus over 3 m of sinking (SKG-CAL-001 I6). The pipe, drum and bentonite belong to the trial kit, not the BOM. No figure for the lubricated friction is claimed: the string still sinks only while it stays below 1.66 kPa, and the test pit must show it.
2. **O3, option A.** PR-012 and PR-013 are accepted as written. PR-012 is met by design (K5, K22); PR-013 sets what the test pits must show.
3. **O4, option C.** The crank shear pin and the shear link are removed. - *Slip clutch (BOM line 24).* The 12-tooth sprocket and two 6 mm driven discs, 84 mm across, are welded to a 36 mm sleeve that runs on the crank shaft.

A woven facing 84 x 44 x 4 mm bonded to an 8 mm inner flange welded to the shaft, and another bonded to an 8 mm pressure plate on an 8 x 7 mm key, grip the discs, clamped by two disc springs 50 x 25.4 x 2.0 in series and a KM5 locknut with an MB5 tab washer. It is set on CalRig to slip at about 34 N m, which caps the capstan pull at 1,472 N (the tripod's 150 kg rating) on the first rope layer at the high end of a ±5 % setting (SKG-CAL-001 C1 to C3). The chain guard grows 21 mm outboard and 11 mm inboard to enclose it, so the -X crank and the shaft end move out 21 mm. - *Load indicator (BOM line 14).* On the crosshead's dead-end lug, two 6 mm straps on a 16 x 42 mm clevis pin carry a housing plate; the line's thimble sits on a fork that draws an M8 rod up through the housing against 14 disc springs 18 x 9.2 x 1.0. The rod travels about 2.8 mm at 1.47 kN; rings on the rod, set on CalRig in 0.25 kN steps, and an O-ring tell-tale show the peak of each lift. It does not release. - *Proof basis.* PR-007 is restated: proof at 1.25 times the rated load of the line, the clutch's highest slip (150 kg), so 188 kg, inside the HatchSide proof of 225 kg, with the clutch locked out for the test; the clutch is set and checked before each campaign. Which lifting standard applies stays to assess (SKG-REG-001).

4. **O5, B (after checking A).** The 2.0 m T-bar's ends sweep 1,017 mm and clear the HatchSide legs only below 801 mm above ground, while the doors stand at 390 mm and the T-bar moves 2 m with each pole section (SKG-CAL-001 H10), so the layout does not allow it. Four people turn the scraper, two on each side of the 1.4 m T-bar: 71 to 84 N each (H9). No hardware changes.

Consequences

- **Requirements.** PR-007 restated and at risk: the window between the steady full-grab pull at the capstan (1,068 N) and the clutch's lowest slip (1,127 N, third layer) is 6 %, on a ±5 % setting only CalRig can show. PR-001 stays not met unaided and is at risk with the annulus. PR-012 and PR-013 are no longer proposed.
- **STEM check.** S4 to S7, S10, S10a, S15, S16, S18, S21, E2, B4 and OB2 restated (SKG-STM-001): S6, S7, E2, S21 and B4 move from failing to plausible, to confirm at TRL 4; the check is still open with OB5 and OB6 (option B) failing.
- **Mass and cost.** The grab is 52.7 kg (52.5 before); the capstan 71.1 kg. Value-engineering target: USD 4,000. Estimated cost of the constructable design: USD 3,063 (USD 937 under the target). The decisions add USD 77, all estimates: clutch USD 60 in place of USD 8 of shear pins, load indicator USD 35 in place of the USD 18 link, crank shaft USD 5, chain guard USD 3. `budget_usd` unchanged.
- **Safety.** A snag now stops the hoist at the clutch, with the pawl holding the drum, instead of opening the grab. The clutch must never be tightened past its CalRig setting: the locknut is paint-marked and the load indicator shows every lift's peak, so a clutch that has crept high is seen. Four people at the T-bar stand clear of the open doors.
- **Still open.** O1, concept A or B (no pick, within 10 %).